



SPACE HOTEL

In the 1990s the Space Island Group produced a plan for a wheel-shaped orbital space hotel that would be constructed out of used Space Shuttle fuel tanks. If suborbital space trips for paying passengers become commonplace, there will inevitably be demand for longer stays in space.

orbiting space station. If space tourism took this direction, then the space station would presumably evolve into a space hotel where the customers would spend an interestingly unusual vacation with spectacular views.

They would necessarily be ultra-rich individuals, with time and a fortune to spare. The limited scale of the operation made it likely that tourism would continue merely to “piggy-back” upon government programs.

The Ansari X-Prize

This vision of the future of space tourism was overturned by the successful development of the world’s first privately financed space vehicle. The leap to private space travel was stimulated by a device that had proved spectacularly successful in the early days of flight: the offer of a significant money prize. Just like Blériot’s flight across the English Channel in 1909 and Lindbergh’s transatlantic flight in 1927, the first space flight by a privately financed reusable vehicle was to be rewarded in cash. In 1996 the X Foundation offered \$10 million for anyone who could reach suborbital

space – defined as above 60 miles (100 km) altitude – twice in the same craft within two weeks. A deadline was set of January 1, 2005.

Twenty-six hopeful teams decided to enter for the prize. Argentina, Britain, Canada, Israel, Romania, and Russia were represented, as well as the United States. In the end it was the American designer Burt Rutan who struck the right blend of innovation and feasibility. Focusing on the key issue of safety, Rutan identified the two points in space flight that carried maximum risk: ground launch and re-entry to the atmosphere. To avoid the first problem, he planned to have his spaceship carried to high altitude under the belly of an aircraft. To cope with re-entry, he equipped the wings with folding booms that would slow the craft through a high-drag “shuttlecock” effect as it plunged toward the Earth, avoiding the extremely high temperatures experienced by traditional spacecraft entering the atmosphere. There would also be no need to strike the atmosphere at a precisely calculated angle.

SpaceShipOne

Rutan’s airborne launch vehicle, the White Knight, made its first flight in summer 2002. First tests with the three-seater spacecraft, SpaceShipOne (SS1), began the following year. It made its first foray into space on June 21, 2004, with pilot Mike Melvill at the controls. The rocket motor, fueled by an unusual mix of nitrous oxide and rubber, ignited after the spaceship was released from the White Knight into gliding flight. The spaceship then accelerated upward,

Space tourism

The fact that the exploration of space developed in the context of the Cold War competition between the United States and the Soviet Union ensured that manned space flight was promoted by governments rather than by private entrepreneurs, and that its primary goal was national prestige. Although business became an important source of finance for satellite launches, the privatization of manned space travel did not begin until Dennis Tito made his historic flight as the first space tourist in 2001 (see page 372). Tito was able to travel to the ISS because Russia’s post-communist economic debacle had left its space program chronically short of funds. Milking rich businessmen of millions of dollars in return for a trip into space offered a simple and viable way for the Russians to raise cash.

The model of future space tourism suggested by Tito’s trip was complex and expensive. It involved months spent training the paying passengers as cosmonauts and a journey to an

